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Nano/micro-plastics effects in agricultural landscapes: an overlooked threat to pollination, biological pest control, and food security

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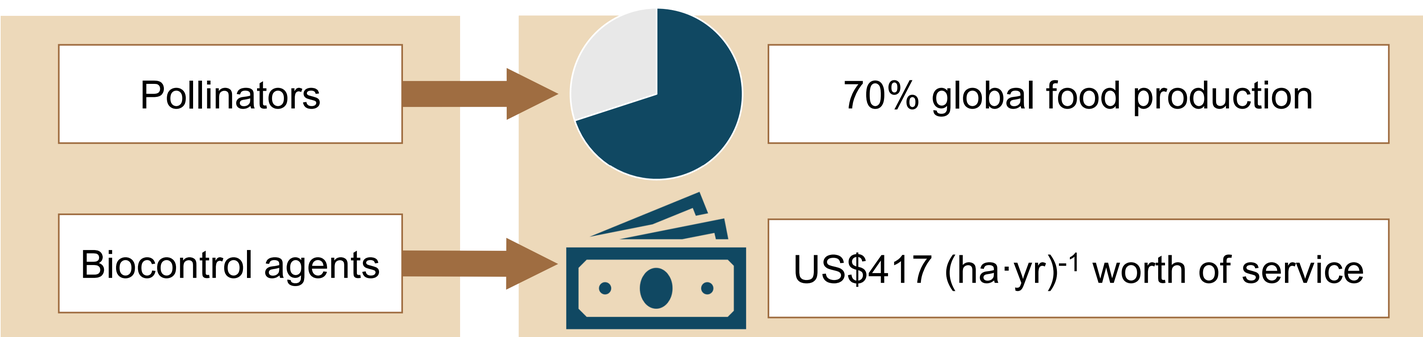


Arthropod ecosystem service providers

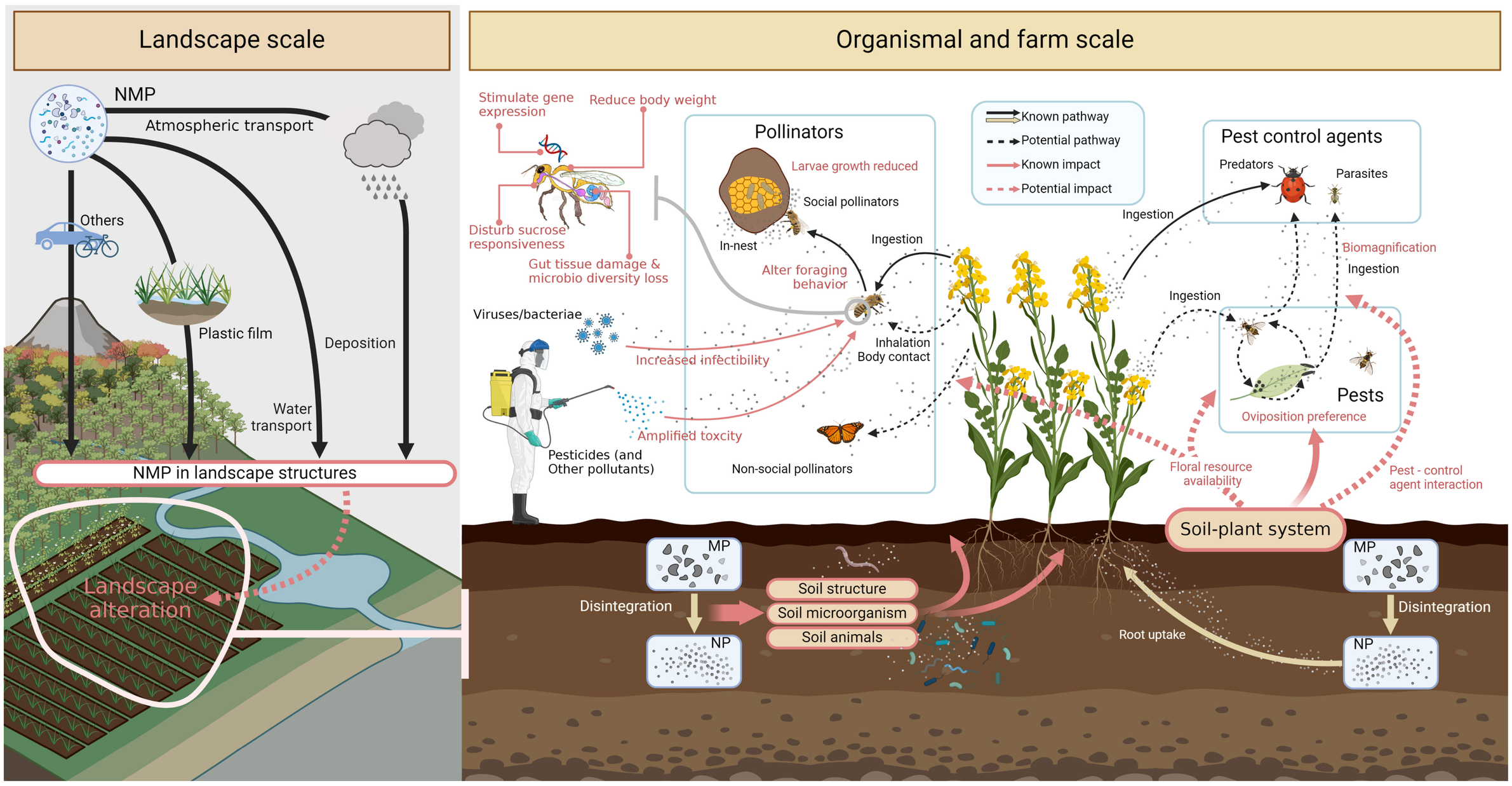
Sustainable agriculture & food security

Nano/micro-plastics (NMP) > plastic particles with diameter ≤ 1 μm, or 1 μm – 5 mm

Pollinators > transfer pollen between flowers

Biocontrol agents > organisms that help resist agricultural pest population

Cross-scale NMP effects on pollinators and biocontrol agents



Roadmap for future researches

Methods to detect NMP in environmental samples.

- Detection of small-sized plastics in real environments and in organisms remains
- Methods to automate sample preparation in NMP research, especially in terrestrial systems
- Standard methods to effectively track NMP and their interaction effects in vivo

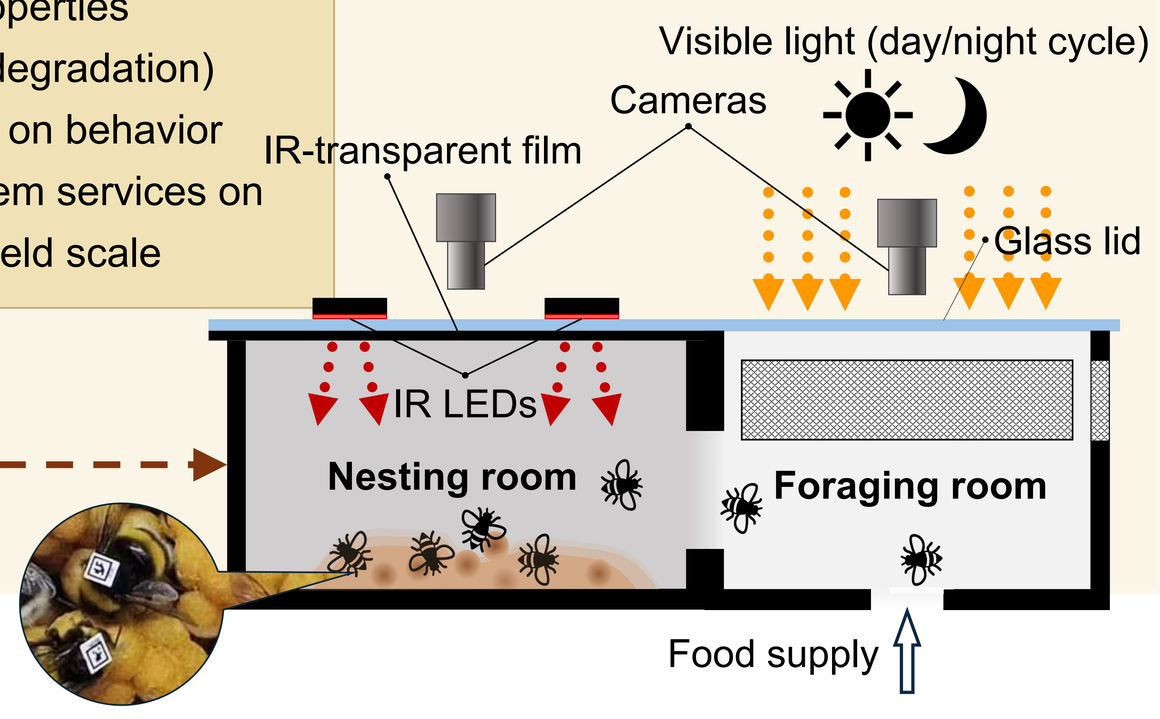
Ecosystem Ecotoxicology of NMP.

- Ecotoxicity of NMP on pollinators and biocontrol agents
- Realistic exposure concentrations Systematic perspectives involving various ecosystem agents
- Realistic NMP properties (weathering and degradation)
- Quantified effects on behavior and their ecosystem services on the semi-field to field scale

NMP impact mitigation.

- Vegetation barriers for airborne NMP transfer
- Mitigation by landscape heterogeneity
- … …

On-going project: microplastic effects on bumblebee *(Bombus terrestris)* individual behaviors and colonial development, base on computer-vision, object detection and miniature matrix code detection.



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